

— A PROGRESS REPORT

Finding the Edge

Dense Neural Networks & TabPFN for NBA Win Prediction

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Project Roadmap

From raw API calls to tested models to live betting edge.



01

Data Ingestion

Web scrape NBA API; ~30 bulk calls vs 12,000+

02

Feature Engineering

Rolling stats, missing-player impact, form

03

Model Testing

Dense NN + TabPFN for tabular data

04

Diagnostics

Assess accuracy, AUC, spread MAE

05

Line Integration

Cross-ref sportsbooks for best user value

Goal *Beat the sportsbook — or at least know when to sit one out.*

Project Purpose

Core Questions

- 01 *Can recent player-level and team-level data predict NBA outcomes before tipoff?*
- 02 *How does this model of prediction compare against the global bookmakers?*

Why It Matters

Better sports analytics creates new business opportunities across industries and sectors.

Teams

Scouting

Media

Storylines

Fans

Engagement

CORE IDEA Use **structured historical data** to estimate outcome probabilities

What the Model Predicts

Classification Task

Predicts whether the home team wins

0-1

Probability

65.7%

Accuracy

0.721

ROC AUC

Regression Task

Predicts point spread (Home Points - Away Points)

+

Home Favored

-

Away Favored

11.36

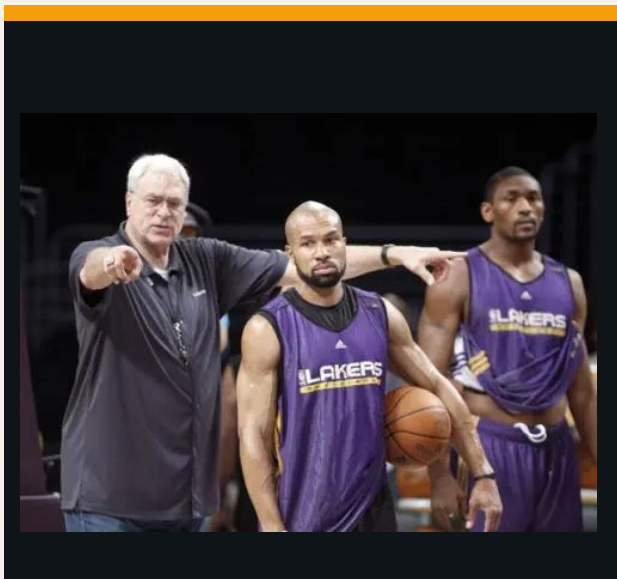
MAE

OUTPUT

Each matchup returns

-  *Win probability*
-  *Predicted spread*
-  *Predicted winner*

A Good Model Requires Good Prep



01

Connect

Hit the NBA Stats API with browser-realistic headers to avoid rate limiting

02

Pull

Grab entire seasons of stats, standings, and game logs in bulk endpoint calls

03

Save

Export four clean CSVs: per-game, advanced, standings, and box scores

Building for a Successful Run



01

Aggregate & Score

131,292 player box scores collapsed into 12,300 team-game rows, with per-game shooting % (FG, 3P, 2P, FT) computed for each team each night.

02

Rolling Form

18 rolling features shifted by 1 game to prevent leakage. PTS, FGM/A, REB, AST, STL, BLK, TOV, shooting %s — plus a Last-10-Wins momentum flag.

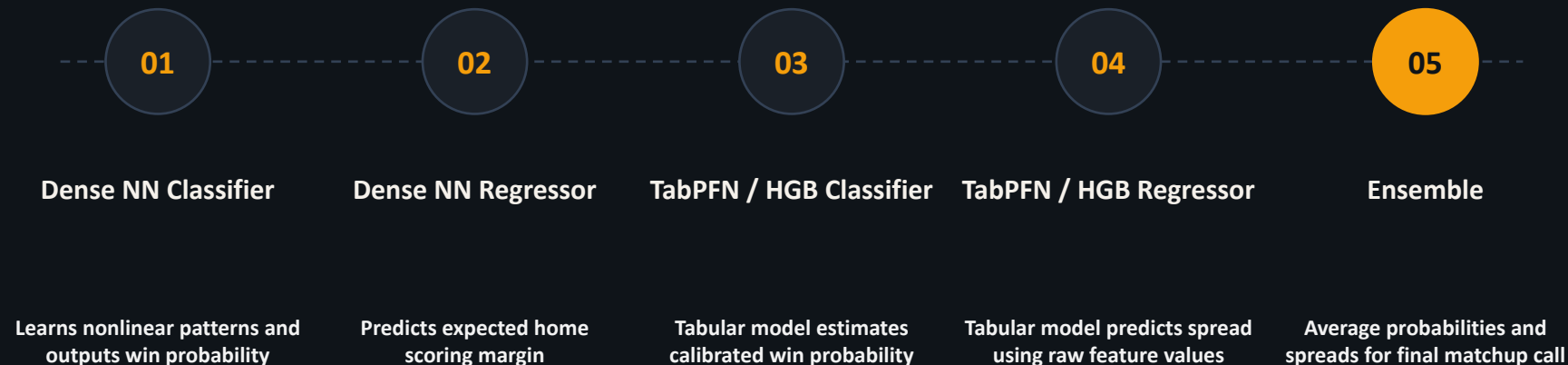
03

Missing Impact

Cross-reference rosters vs. who actually played. Flag games missing key players (USG% ≥ 18 or PPG ≥ 12) and quantify the total missing production.

Modeling Strategy

Two model families & two prediction tasks



Why two models?

Agreement increases confidence; disagreement highlights uncertainty.

Dense Neural Network

What It Is

Captures interactions like recent scoring form + shooting efficiency + missing players

23

Inputs

96-48-24

Hidden Units

8,833

Params

What It Does

Classifier predicts wins; regressor predicts expected point spread

Sigmoid

Win Prob

Linear

Spread

Huber

Reg. Loss

StandardScaler

Neural nets train better when features sit on comparable scales

REGULARIZATION

Dropout + L2

Prevents the model from memorizing and overfitting

Early Stopping

Training stops when validation performance stops improving

TabPFN

What It Is

A tabular foundation model designed for strong predictions on small-to-medium datasets

Tabular

Data Type

5,891

Rows

23

Features

What It Does

Classifier predicts wins, regressor predicts expected point spread

Sigmoid

Win Prob

Linear

Spread

Pre-trained for synthetic data

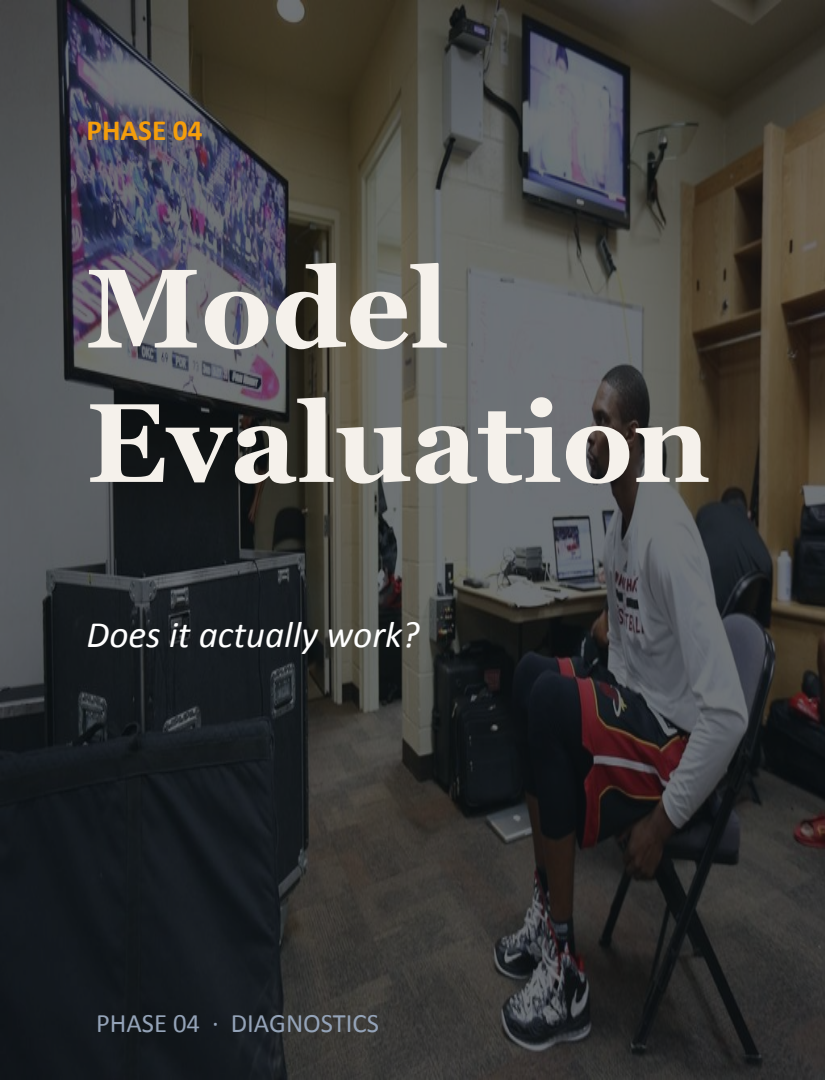
Avoids over-prediction problems with tabular small datasets

HGB Classifier

HistGradientBoosting + sigmoid calibration improves probability usability

HGB Regressor

Robust spread prediction against outliers

A man is sitting in a chair in a room. On the left, a large screen displays a game with the text "PHASE 04" in the top left corner. The man is wearing a white long-sleeved shirt and red and black shorts. He is looking towards the screen. In the background, there is a whiteboard and a desk with a laptop. The room has wooden shelves and a carpeted floor.

PHASE 04

Model Evaluation

Does it actually work?

PHASE 04 · DIAGNOSTICS

BASELINE BEATEN BY

+10 pts

Both models outperform 55.3% home-win rate

DENSE NN LEADS ON AUC

0.721

vs. 0.705 (TabPFN fallback) — edge in signal strength

WIN ACC FROM SPREAD

66.1%

Spread-derived calls beat classifier on directional picks

Performance

Dense NN

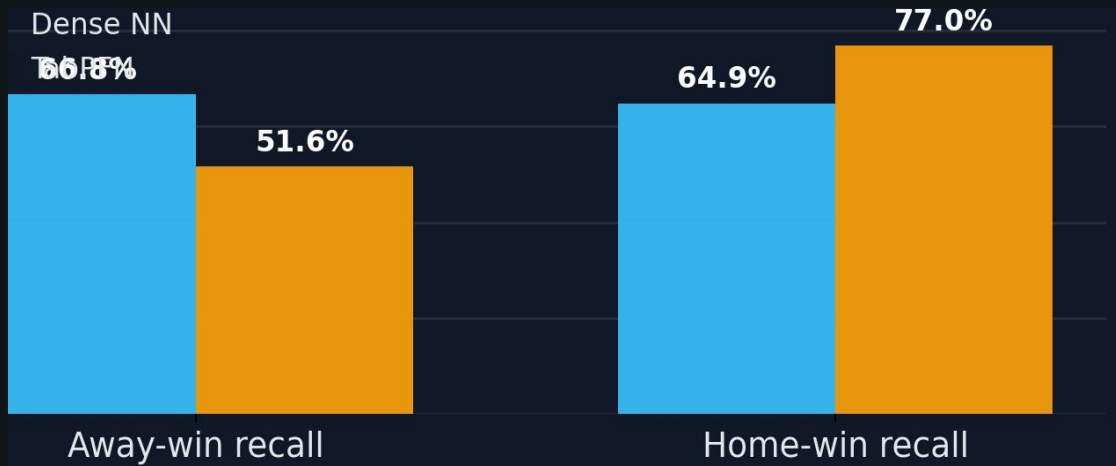
65.7% accuracy

775 correct · 404 missed

TabPFN

65.6% accuracy

774 correct · 405 missed



Game Time: A Multifaceted Plan

Dense Neural Network

Keras · batch-norm, dropout

65.7%

Accuracy

0.721

ROC AUC

11.36

Spread MAE

TabPFN (HGB fallback)

Foundation model for tabular data

65.6%

Accuracy

0.705

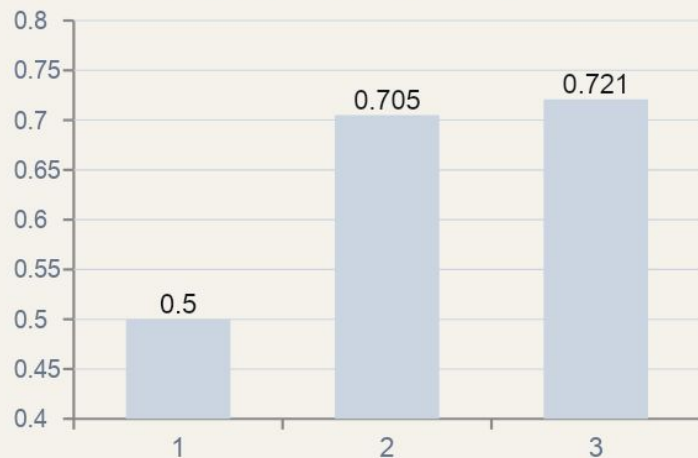
ROC AUC

11.66

Spread MAE

ENSEMBLE BOS→MIA 59.5% · OKC→DAL 71.0% · LAL→GSW 70.5%

ROC AUC on 1,179-game test set





Odds API Integration

How does this model compare against the larger prediction models?

Odds API

JSON Data on real-time and historical betting odds across sports.

Includes H2H (moneyline), spread and totals.

Our Use

Pull moneyline & spread and map to match team names and dates.

Converts odds into probabilities and matches the market predictions to our model.

+

Model predicting win more likely

-

Market predicting win more likely

**Movement
Recommendations**

Call when the edge passes a threshold

PHASE 05

Finding the Edge

Cross-reference model spreads against posted lines to find the best bang for the buck.



MATCHUP	MODEL	BOOK LINE	EDGE	CALL
LAL vs GSW	+7.2	+5.5	+1.7	STRONG
OKC vs DAL	+6.4	+4.0	+2.4	STRONG
BOS vs MIA	+4.0	+5.5	-1.5	FADE
DEN vs MIN	+6.8	+6.0	+0.8	LEAN
NYK vs PHI	+7.6	+4.5	+3.1	STRONG



Looking Forward

Where we're going from here.

MODEL

Expand Features

Incorporate injury reports, rest days, and schedule to better capture conditions

VALIDATION

Real-Time Backtesting

Run a true historical backtest with only the information available at the time

USE CASE

Automation

Create a daily pipeline to run predictions automatically with updated API data

Thank you.

Questions?

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